System Engineering Plan

SQRT (Sub-zero Quantitative Research Team)

Caimin Keavney, Elyana Abdallah, Javier Ayuso, Orlaith Ni Dhuill, Bruce Ritter

Document History

Date	Version	Notes	Approver
2026–02–04	SQT-PDR-SEP-0	Initial release	EA

Author Contributions

CK §1–6

OND §3

JA §5

BR §6

EA §6

EA Technical Reviewer

Contents

1 Executive Summary **3**

2 Mission Overview **4**
2.1 Mission Objective 4
2.2 Success Criteria 4

3 Mission Requirements **4**
3.1 Level 0: Mission Objectives 4
3.2 Level 1: Mission Requirements 5
3.3 Level 2: Technical Requirements 7

4 Function Tree **10**

5 Product Tree **11**

6 System Development Plan **11**
6.1 Design . . . 12
6.2 Resource Procurement . . . 12
6.3 Prototyping . . . 12
6.4 Software Development . . . 13

List of Figures

1 SQRT Function Tree . . . 11
2 SQRT Product Tree . . . 11

List of Tables

1 Customer (UCD) Mission Requirements . . . 5
2 Mission Specific Requirements . . . 6
3 Structural Requirements . . . 7
4 Communications Requirements . . . 7
5 Housekeeping Telemetry & Atmospheric Data . . . 7
6 Power Requirements . . . 8
7 Operations Requirements . . . 8
8 Water Container Requirements . . . 8
9 Exposure Mechanism Requirements . . . 9
10 Thermal Camera Requirements . . . 9
11 Sensors Requirements . . . 10

Key Reference Documents

1. UCD-PRD-2026-0
2. UCD-DRD-2026-0
3. SQT-PRD-MDD-0
4. SQT-PRD-PMP-0
5. SQT-PRD-DJF-0

1 Executive Summary

This document will provide a preliminary overview of the requirements, associated risks and function/product trees regarding the SQRT mission and its development. In Sec. 2, a mission overview will be provided, including mission objectives and metrics of mission success. Further details regarding these topics and the scientific motivation can be found in the Mission Description Document, SQT-PDR-MDD-0.

The system engineering approach to the risks associated with this mission will be covered. The requirements addressing these risks will be defined in Sec. 3, encompassing risks to the scientific objectives, system components, and the overall architecture of the TupperSat. Each Level 1 and Level 2 requirement will be justified with regard to the mission objectives. The customer requirements are primarily concerned with the flight readiness and structural integrity of the TupperSat, while prioritising ground segment communications to facilitate recovery operations. Mission specific requirements will also be defined. These requirements are concerned with the realisation of the science goals and the collection of meaningful data. Mission critical subsystems, such as the exposure mechanism and the low-pressure chamber, are identified and treated accordingly.

The systems, including both processes and resources, required to design a TupperSat that conforms to these requirements and yields meaningful results will be presented visually in sections 4 and 5. The function breakdown will be consulted to appraise the necessity of requirements, while the product tree will be used to define work packages for their design and assembly (SQT-PDR-PMP-0). The system development plan found in Sec. 6 will define the system engineering activities to be performed in phase 2 of the project in preparation for the Critical Design Review. This will primarily consist of designing processes, procurement of necessary components and prototyping of mission-critical subsystems for testing in phase 3.

2 Mission Overview

2.1 Mission Objective

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to answer the question of how water plumes on icy moons thermally evolve.

This investigation into the thermal mechanisms behind cryovolcanic plumes will be done using a high-altitude demonstration. This will involve the sudden exposure of liquid water within the TupperSat to a colder, lower-pressure environment which will be generated by elevating the TupperSat system to an altitude of 20-25 km. At 25 km altitude, the average atmospheric pressure is about 25 mbar (approx. 2.5% of that at sea level) with a corresponding boiling point of water of approx. $21°C$. To perform this exposure, an opening mechanism, consisting of a valve, a trigger sensor, and a servo motor, will allow a small volume of water to "see" a colder, less pressured environment within the confines of the TupperSat.

The mission objectives must be clearly stipulated in order to define a set of relevant requirements. Requirements will be reviewed and appraised based on their necessity and their contribution to the fulfillment of the mission objectives. These mission objectives and the corresponding mission and technical requirements will be detailed in Sec. 3.

2.2 Success Criteria

The success of the SQRT mission will be determined by a number of metrics. Firstly, the flash evaporation event will have to be detected and verified. In this phase, the current idea is to perform direct observation via use of a thermal camera. The rapid phase transition will hopefully be decipherable and interpretable via analysis of the thermal evolution of the water. The footage or frames acquired by this camera will be stored onboard the TupperSat by means of an SD card. In the event of non-retrieval of the TupperSat, a number of non-direct, less data intensive verification methods are being explored. A small thermocouple or temperature sensor could be placed inside the low-pressure chamber to confirm a rise in temperature due to the introduction of the relatively warm water. A preliminary metric of mission success will be the activation of the exposure mechanism and its deployment. Successful activation of the altitude trigger sensor at the chosen target altitude will be signified by the sending of a signal. Fulfilling these criteria and maximising mission return will require the design of a lightweight system that is economical with the data downlink and is compartmentalised to prevent water damage to electronics. The continued design of this system in phase 2 will be guided by the mission & technical requirements outlined in this document.

3 Mission Requirements

The mission requirements stipulated below are divided into a number of sections. Level 0 requirements correspond to the mission objectives, against which the success of the mission will be measured. Level 1 requirements are high-level mission requirements directly linked to the mission objective, data acquisition, and retrieval. The level 2 requirements are technical requirements concerned with the general operation of the TupperSat platform and structure. These requirements will also be documented and traced by means of a requirement traceability matrix during the successive phases of the mission.

3.1 Level 0: Mission Objectives

MO1 Fly and recover a TupperSat containing an experimental scientific or engineering/technology demonstration payload with potential application to space science.

MO2 SQRT will attempt to reproduce and observe the flash evaporation process of water exposed to near-space conditions.

3.2 Level 1: Mission Requirements

3.2.1 Customer (UCD) Requirements

Table 1: Customer (UCD) Mission Requirements

Requirement ID	Title	Description	Justification
UCD:MR:0010	Payload Size	The TupperSat shall be carried as a payload on a light unmanned free balloon.	Inherited from UCD-PRD-2026-0.
UCD:MR:0020	Flight Time	The TupperSat shall operate for at least 6 hours.	Inherited from UCD-PRD-2026-0.
UCD:MR:0030	Flight Environment	The TupperSat shall operate up to 35 km altitude.	Inherited from UCD-PRD-2026-0.
UCD:MR:0040	TDRS Communication	The TupperSat shall communicate with a ground station using TDRS.	Inherited from UCD-PRD-2026-0.
UCD:MR:0050	Tracking and Recovery	The TupperSat shall report real-time position telemetry.	Inherited from UCD-PRD-2026-0.
UCD:MR:0060	Non-Recovery	The TupperSat shall transmit any experimental data necessary to meet the mission objectives in the event that recovery is not possible.	Inherited from UCD-PRD-2026-0.
UCD:MR:0070	Atmospheric Profiling	The TupperSat shall measure and report the atmosphere's temperature and pressure profiled with vertical resolution of at least 1 measurement per 200 m ascent (goal: at least 1 measurement per 100 m).	Inherited from UCD-PRD-2026-0.
UCD:MR:0080	Launch Readiness	The TupperSat shall be deilvered ready to be turned on for launch without further intervention.	Inherited from UCD-PRD-2026-0.

3.2.2 Mission Specific Requirements

Table 2: Mission Specific Requirements

Requirement ID	Title	Description	Justification
SQT:MR:010	Thermal Data	The TupperSat shall measure the temperature of the water and/or surrounding vapour during exposure to reduced pressure	To characterise and quantify its thermal evolution and identify key transition events.
SQT:MR:020	Pressure Data	The TupperSat shall measure the pressure within the low-pressure chamber throughout the flight.	The pressure readings collected will allow for analysis of the waters thermal evolution and its conformance to expectations.
SQT:MR:030	Timestamped Data	The TupperSat shall record the time at each primary measurement.	Storing timestamps at each data acquisition will facilitate chronological results.
SQT:MR:040	Altitude Data	The TupperSat shall record the altitude for all primary data acquisitions.	Recording the altitude will verify if the trigger sensor and exposure mechanism functioned at the desired altitude.
SQT:MR:050	Trigger Operation	The TupperSat shall provide verifiable confirmation that the trigger condition has been met	This requirement supports UCD:MR:0060 for the scenario of non-recovery of the TupperSat.
SQT:MR:060	Exposed Sample Containment	The TupperSat shall maintain the water sample within the low-pressure chamber for a time sufficient to observe thermal evolution after exposure.	This requirement ensures the acquisition of sufficient data for thermal evolution studies. The specific time can be defined once the camera is procured and the frame rate is a known variable.
SQT:MR:070	External Environment Readings	The TupperSat shall record continuous external pressure data during the exposure period.	This requirement will aid post-mission analysis. External pressure can be compared to the pressures inside the low-pressure chamber to appraise performance of venting.
SQT:MR:080	Ease of Data Retrieval	The TupperSat shall be designed in such a way that primary experimental data is recorded, stored and retrievable following payload recovery	For post-mission analysis and subsequent confirmation of mission success.
SQT:MR:090	Viability for Prototyping	The TupperSat shall be designed to allow repeatable ground-based testing	To ensure reliability and durability of mission-critical subsystems.
SQT:MR:100	Water Exposure	The TupperSat shall be designed in such a way to expose liquid at approx. 25 km in altitude.	This requirement helps to satisfy MO2 and to ensure that flash evaporation occurs.

3.3 Level 2: Technical Requirements

3.3.1 Customer (UCD) Requirements

3.3.1.1 Structure

Table 3: Structural Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00010	TupperSat Mass	The TupperSat's total mass shall not exceed 0.8 kg.	Inherited from UCD-PRD-2026-0.
UCD:TR:00020	TupperSat Volume	The TupperSat shall be enclosed in a Tupperware-like container, with volume less than 2 litres.	Inherited from UCD-PRD-2026-0.
UCD:TR:00030	Launch Interface	The TupperSat shall attach to the launch balloon using a single carabiner connector.	Inherited from UCD-PRD-2026-0.

3.3.1.2 - Communications

Table 4: Communications Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00100	Packet Transmission Frequency	The TupperSat shall transmit packets at 868 MHz.	Inherited from UCD-PRD-2026-0.
UCD:TR:00110	Packet Drop Rate	The TupperSat shall achieve a packet drop rate of less than 20% transmitting over a distance of at least 12 km.	Inherited from UCD-PRD-2026-0.
UCD:TR:00120	Telemetry Packet Rate	The TupperSat shall transmit at least 2 and no more than 3 housekeeping telemetry packets per minute throughout the flight duration.	Inherited from UCD-PRD-2026-0.
UCD:TR:00130	Data Packet Rate	The TupperSat shall transmit no more than 3 data packets per minute throughout the flight duration.	Inherited from UCD-PRD-2026-0.

3.3.1.3 Housekeeping Telemetry & Atmospheric Data

Table 5: Housekeeping Telemetry & Atmospheric Data

Requirement ID	Title	Description	Justification
UCD:TR:00200	Telemetry - Altitude	The TupperSat shall record its altitude at least every 10 s with accuracy better than ± 50 m.	Inherited from UCD-PRD-2026-0.
UCD:TR:00210	Telemetry - Latitude & Longitude	The TupperSat shall record horizontal position (latitude and longitude) with horizontal accuracy better than 50 m.	Inherited from UCD-PRD-2026-0.

Requirement ID	Title	Description	Justification
UCD:TR:00220	Telemetry - Internal Temperature	The TupperSat shall record internal temperature with accuracy better than $\pm 2^{\circ}C$ at least every 20 s.	Inherited from UCD-PRD-2026-0.
UCD:TR:00230	Telemetry - External Temperature	The TupperSat shall record external temperature with accuracy better than $\pm 2^{\circ}C$ at least every 20 s.	Inherited from UCD-PRD-2026-0.
UCD:TR:00240	Telemetry - Atmospheric Pressure	The TupperSat shall record atmospheric pressure with accuracy better than 20 mbar at sea level at least every 20 s.	Inherited from UCD-PRD-2026-0.

3.3.1.4 Power

Table 6: Power Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00300	TupperSat Power	The TupperSat battery system shall provide power for at least 6 hour of flight operations.	Inherited from UCD-PRD-2026-0.

3.3.1.5 Operations

Table 7: Operations Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00400	Delivery	The TupperSat shall be delivered to the launch site ready to be launched by a single third-party operator in under 15 minutes	Inherited from UCD-PRD-2026-0 & quantifies UCD:MR:0080

3.3.2 Mission Specific Requirements

3.3.2.1 Water Container

Table 8: Water Container Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00100	Water Temperature	The water container shall maintain the water at a temperature over $25^{\circ}C$.	At 25 km altitude, the atmospheric pressure will be around 25 mbar. At this pressure, water boils at around $21^{\circ}C$.
SQT:TR:00110	Pressure Resistant	The water container shall maintain structural integrity (no cracks or leaks) during ascent. This includes all pressure differentials that may be encountered.	This requirement ensures containment of water and prevention of sensor or electronic damage.

3.3.2.2 Exposure Mechanism

Table 9: Exposure Mechanism Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00200	Valve Activation	The exposure mechanism shall send a signal after operation for downlink to ground segment.	This requirement will confirm that the exposure was executed. Satisfies SQT:MR:050 for event of non-recovery.
SQT:TR:00210	Leak Avoidance	The exposure mechanism shall prevent premature water exposure at environmental conditions up to the desired altitude.	This requirement is to prevent damage to electronics and to optimise data acquisition.
SQT:TR:00220	Valve Operation	The exposure mechanism shall operate at low-pressure chamber temperatures down to $-40^{\circ}C$.	This requirement ensures that the exposure will occur, satisfying MO2 by allowing for the acquisition of meaningful thermal data.
SQT:TR:00230	Tubing Material	The tubing used for passing water into the chamber shall remain operable (allow water to pass through) at temperatures as low as $-40^{\circ}C$.	Similarly to SQT:TR:00220, this requirement will ensure that water will reach the low-pressure chamber for thermal observations. Tubing will be more vulnerable than rest of exposure mechanism, justifying a dedicated requirement.

3.3.2.3 Thermal Camera: Primary Data Acquisition

Table 10: Thermal Camera Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00300	Thermal Resolution	The thermal camera shall have a thermal resolution of $\leq 0.1\,^{\circ}\mathrm{C}$	This requirement helps to confirm the occurrence of a flash-evaporation event. The selected thermal resolution chosen is at least an order of magnitude smaller than the temperature change caused by evaporative cooling during flash evaporation
SQT:TR:00310	Camera Function	Thermal camera shall remain functional in temperatures down to -10 $^{\circ}C$.	This requirement will allow for measurement taking in ventilated, low-pressure chamber. The camera will have to be insulated or passively heated in order to remain functional.

Requirement ID	Title	Description	Justification
SQT:TR:00320	Frame Rate	Thermal camera shall have a frame rate of over 30 frames per second	Flash-evaporation occurs in less than a second, necessitating a sufficient frame rate to observe the process, as opposed to recording only pre- and post-event temperatures.
SQT:TR:00330	Calibration Stability	The thermal camera shall maintain calibration accuracy over the expected flight temperature range.	Ensures validity of recorded thermal data.
SQT:TR:00340	Field-of-View	The thermal camera shall have an FOV of at least 15 degrees.	This requirement ensures that, with current preliminary designs, the entire exposure mechanism can be seen by the thermal camera. This supports SQT:MR:010.

3.3.2.4 Sensors: Secondary Data Acquisition

Table 11: Sensors Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00400	Chamber Pressure Resolution	The pressure sensor in the low-pressure chamber shall have an accuracy of less than 5 mbar.	The pressure in the chamber at exposure is predicted to be ≈ 50 mbar. Allowance for maximum 10 % inaccuracy.
SQT:TR:00410	Temperature Compatibility	The pressure sensor should operate at temperatures down to $-40°C$.	This ensures that the pressure sensor will provide reliable data from the low-pressure chamber.
SQT:TR:00500	Altitude Sensor Update Rate	The altitude sensor shall update the altitude every 10 s.	This conforms with UCD:TR:00200 and supports experimental triggering within 500 metres of the desired altitude mark.

4 Function Tree

The function tree below defines the processes performed by each subsystem comprising the experimental payload, the TupperSat platform, and the ground segment. Fulfillment of the mission objectives depends on the correct execution and coordination of these functions, ensuring that the exposure event is reliably triggered, observed, recorded, and transmitted for subsequent analysis.

From the function tree, mission critical processes can be identified. One such process is the exposure of the water to the low-pressure chamber at the designated altitude. This function is not only vital to mission returns, but is risk-loaded, combining the operation of a sensor, a motor and the valve. The function tree allows the design phase to be organized and carried out in a productive manner. Together with the product tree, it defines how each component will be expected to perform, and the inter-system communications that will be necessary.

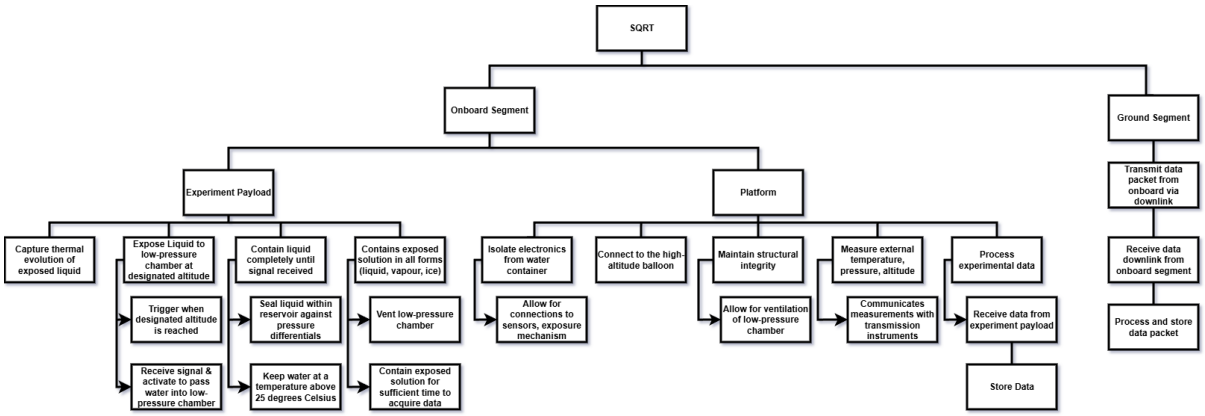


Figure 1: SQRT Function Tree

The development of each of these systems will be broken down into corresponding work packages which will be detailed in SQT-PDR-PMP-0.

5 Product Tree

The product tree below details the products required to perform the functions defined in the function tree in Sec. 4. The specific product name or code is also included, where available. The procurement of components and consumables in phase 2 will be guided by this schematic.

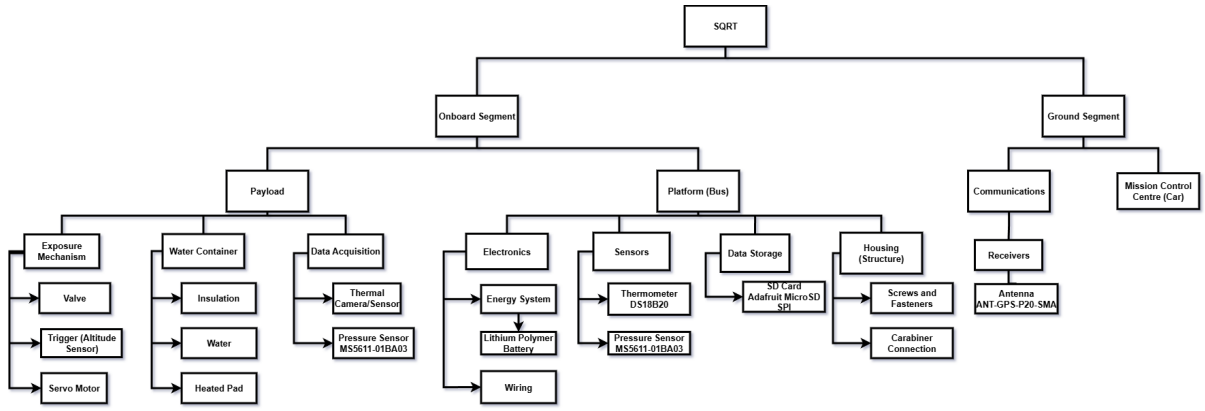


Figure 2: SQRT Product Tree

6 System Development Plan

Operations in phase 2 will work towards the detailed design of each sub-system via the completion of defined work packages found in SQT-PDR-PMP-0. This phase will take place over 3 weeks, culminating in the submission of the Critical Design Review. Acceptance of this review will set the TupperSat design before launch models are assembled and tested.

6.1 Design

Week 2-6
Based on the feedback from the PDR, adjustments will be made to the design. Throughout phase 2, refinements will be made to the design, owing to resource availability, testing capability, and ease of access for sensors and data storage. Modifications to the design could also stem from the need to passively heat the water container using waste heat from the electronics. Thermal analysis will need to be conducted to explore these options.

The design of the exposure mechanism will be another dynamic process, likely spanning multiple iterations. Prototyping will have to be prioritised for this subsystem, with early testing and troubleshooting being a pressing matter. Possible issues that could arise with this mechanism will be discussed further in SQT-PDR-DJF-0.

The evolution of the design will also be concerned with optimising accessibility for certain components. Mission requirement SQT:MR:090 identifies experimental data storage as one of these components. In the event of TupperSat recovery, it is critical that the data storage (SD card) can be easily retrieved. However, this accessibility should not come at the expense of its protection.

6.2 Resource Procurement

Week 4-5
Upon completion and acceptance of the PDR, work will begin to implement the proposed design. First, research will be conducted into the components and consumables that can be sourced in-house, and will therefore incur no financial toll, and those that will need to be procured from external vendors. The procurement process will consist of market research, product identification and request documentation. Market research will be conducted to find economically efficient options that fulfill the technical requirements. The preliminary budget will be consulted to ensure that the €120 cap is not exceeded. Where possible, components will be purchased from one of UCD's standard suppliers (Mouser, Radionics, Farnell, Digikey).

The procurement phase will coincide with the design process, as the design drives the need for certain specifications (mass, energy consumption, temperature ranges, performance). The latter end of the procurement phase will run alongside the prototyping campaign. This will depend on the availability of components relating to the prototypes.

Examples of components that may need to be externally sourced include the thermal camera, servo motor, and water container (syringe).

6.3 Prototyping

Week 5-7
Works will commence on the production of prototypes for the mission-critical mechanisms such as the trigger-motor-valve subsystem required for introducing the water to the low-pressure environment. Emphasis will be placed on early-stage prototyping and testing to validate functionality, identify unforeseen complexities, and reduce development risk prior to final design implementation.

6.4 Software Development

Week 5-11
Software development activities will focus on the implementation and testing of flight code for sensor interfacing, data acquisition, valve control, and thermal camera operation. Emphasis will be placed on robustness, fault tolerance, and reliable operation under near-space environmental conditions. Software validation will be conducted through bench testing and environmental simulations later where feasible.

Overall, the steps taken in phase 2 will serve to prepare the design for a critical review that will appraise its readiness for assembly and testing.